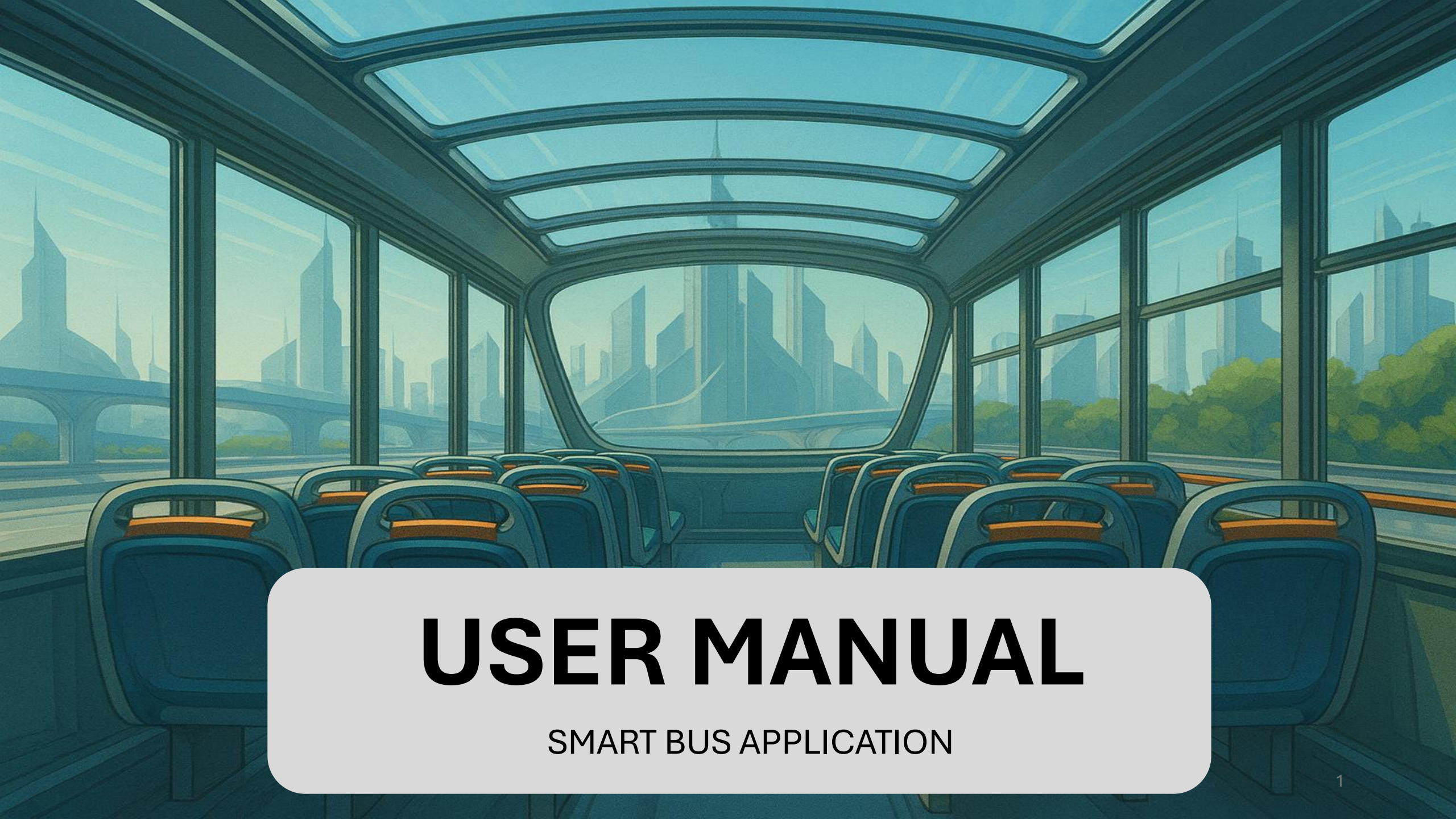

USER MANUAL

SMART BUS APPLICATION

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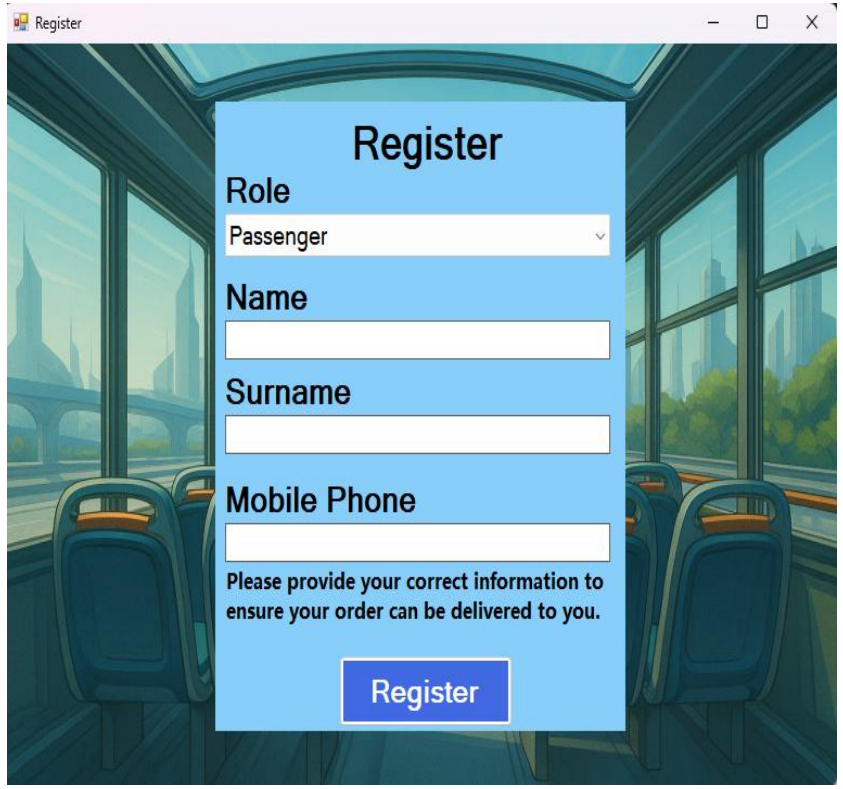
GETTING STARTED

What is it About

- The Getting Started section is designed to guide you through the initial setup and basic navigation of the application.
- It begins with the registration process, providing specific instructions for both passengers and employees to ensure you have the correct access rights and profile settings based on your role.
- Following registration, this section offers an overview of the main screen to help you become familiar with the primary interface and where to find key features.
- Finally, it outlines the proper steps for exiting the application to ensure your session is closed securely and nobody will use your information.

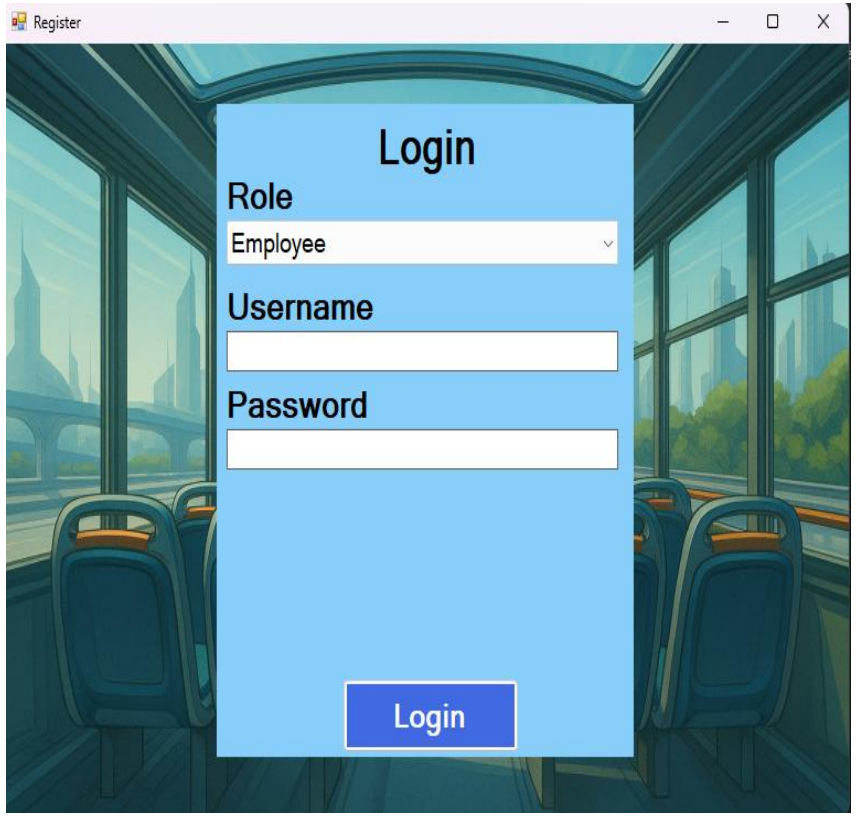
Register as a Passenger

- The registration screen is the first step to accessing the application's various systems and services.
- You must first set the Role selection to Passenger using the dropdown menu.
- Next, fill in the Name and Surname fields with your personal information.
- You are required to provide a correct Mobile Phone number.
- Providing correct information is essential to ensure that any orders you place can be successfully delivered to you.
- Once all fields are completed, click the Register button to finalize your profile and proceed to the main screen.



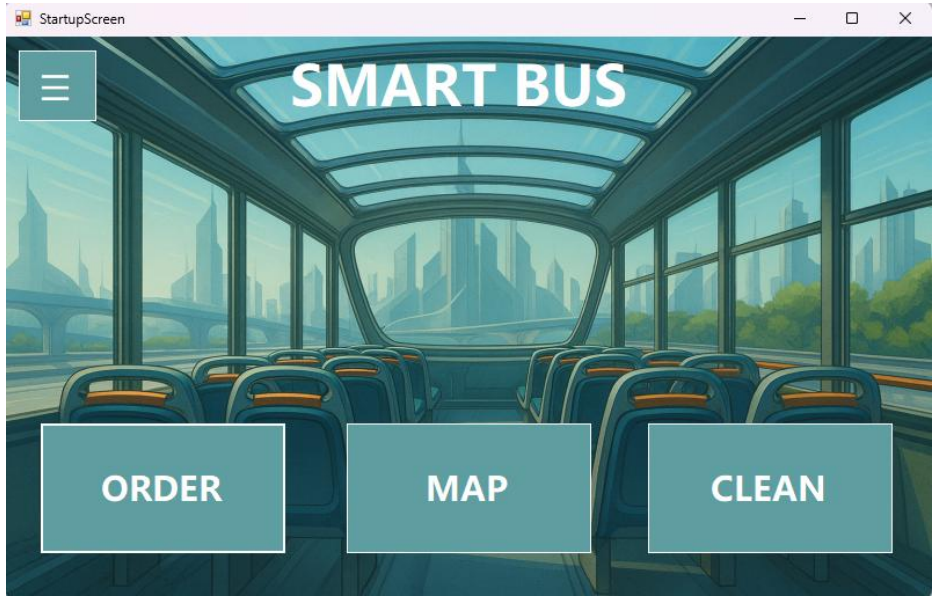
Login as an Employee

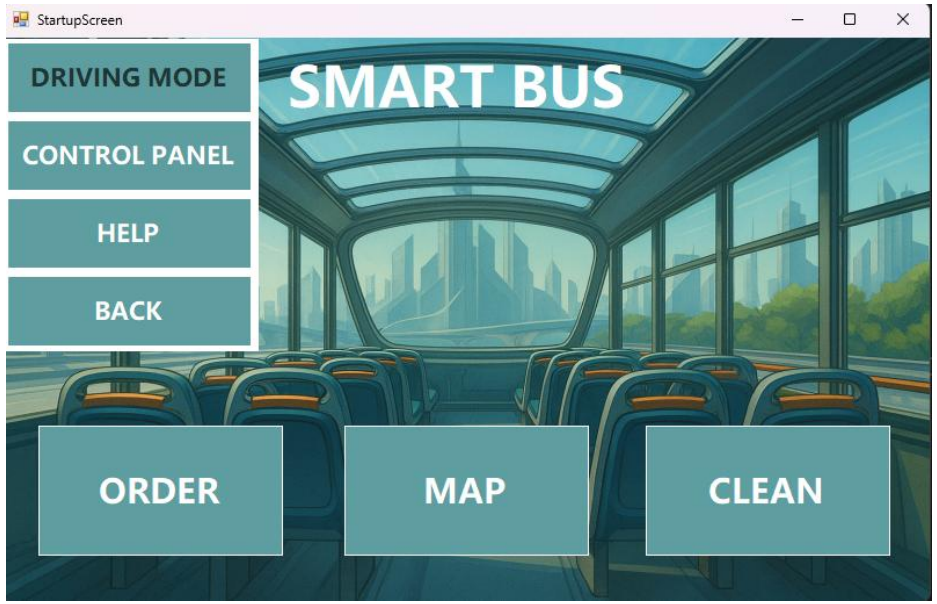
- The Login screen allows staff members to access their administrative features.
- Ensure that the Role selection is set to Employee using the dropdown menu.
- Enter your Username in the Username Field.
- Provide your personal code in the Password field.
- Once your credentials are entered, click the Login button to proceed to the main screen.



Passenger Main Screen Overview

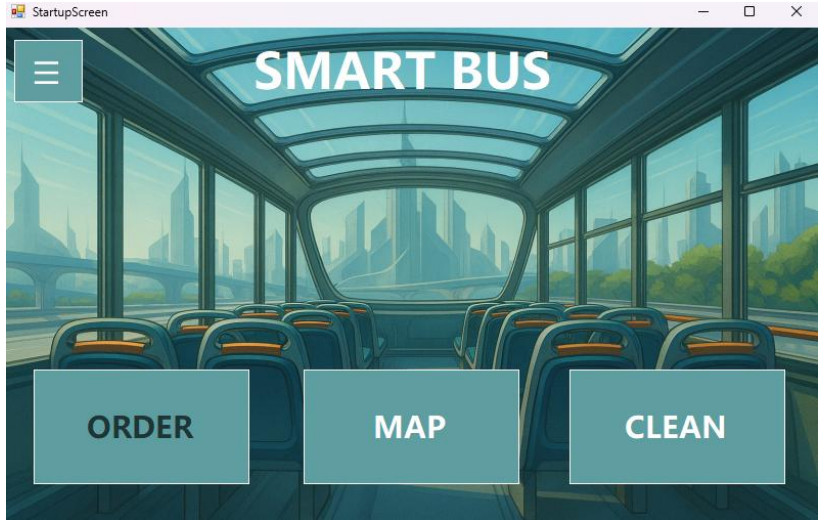
- The passenger interface focuses on travel comfort and on-board services.
- You have full access to the ORDER button to browse the cafeteria and purchase items.
- Use the MAP button to view your current location, bus stops, landmarks along the route and use the driver's view cameras.
- Use the CLEAN button to request cleaning services or view the status of different areas.
- Use the Menu Button(Top Left) to view the Control Panel which shows the A/C temperature, A/C Mode and Energy of the Smart Bus
- Note that the Driving Mode is not available to passengers.
- In the Control Panel you cannot change anything due to safety reasons.
- Use the HELP button to open the User manual.
- Use the BACK button to return to the previous navigation menu at any time.

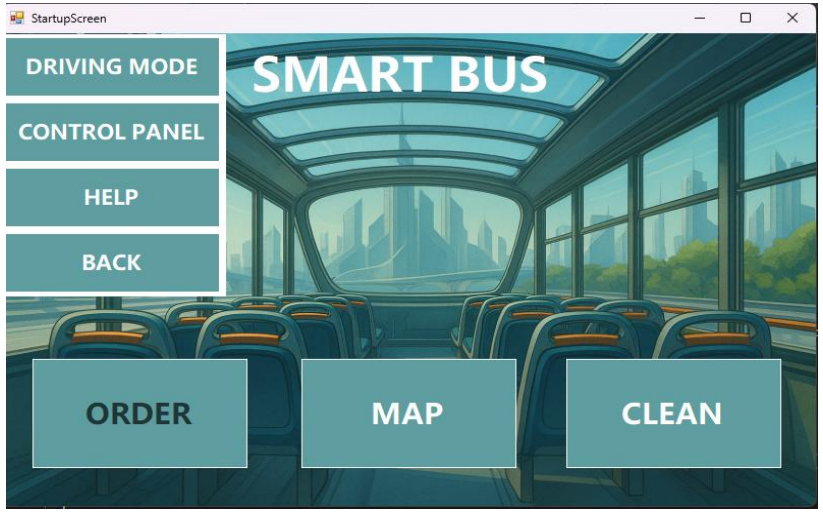




Employee Main Screen Overview

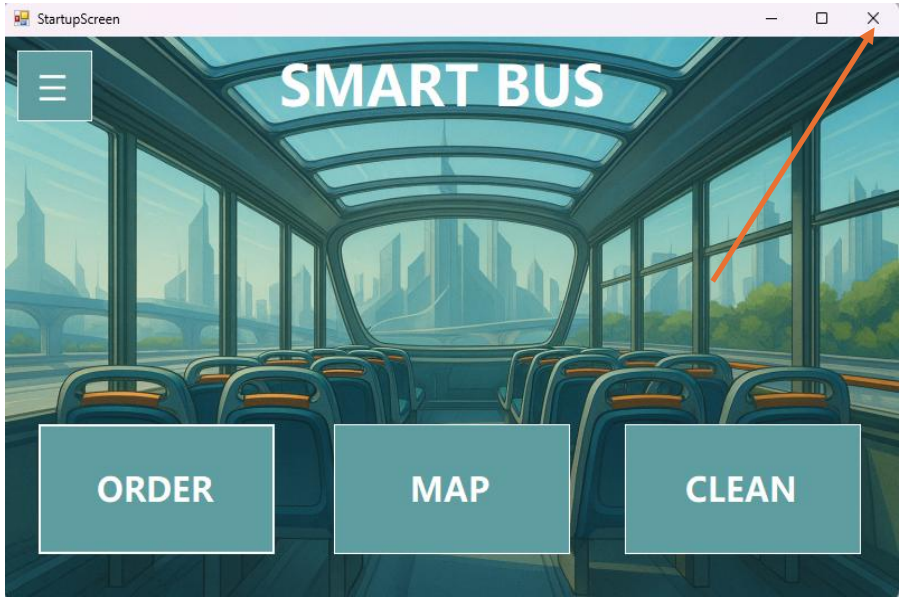
- The employee interface is designed for vehicle management, operational tasks and on-board services.
- You have exclusive access to DRIVING MODE to view the vehicle's speed and get assisted in Driving.
- The CONTROL PANEL allows you to manage technical settings like A/C modes and the bus roof.
- Use the MAP button to view your current location, bus stops, landmarks along the route and use the driver's view cameras.
- Use the CLEAN button to request cleaning services or view the status of different areas.
- The ORDER menu is deactivated in this view because employees cannot place food orders while on duty.
- Use the HELP button to open the User manual.
- Use the BACK button to return to the previous navigation menu at any time.





Exiting the Application

- To return to the previous screen from any form of the application, use the BACK button or the X Button located in the top right corner of every form.
- Once you have finished your tasks, you must return to the main screen to exit.
- To close the application entirely, you must use the X button located in the top right corner of the window.
- Closing the application this way is essential to ensure your session is ended securely.
- This procedure protects your personal information and prevents unauthorized access to your account.



BRIEF GETTING-STARTED NOTES

What is it About

- This section serves as a high-speed guide for users who are interacting with the system for the first time.
- It allows for immediate familiarity with the application, giving the user the opportunity to start testing various features without delay.
- It provides the essential "key steps" needed to operate the main systems in the Smart Bus Application.
- The goal is to help the user understand the basic workflow of the Smart Bus systems in a very short amount of time.

Ordering System

1. Press the ORDER button
2. Choose your desired cafeteria
3. Add the items you want to buy
4. Press the CART button to view your items
5. Choose delivery stop
6. Fill your card information
7. Press the PAY button
8. Wait for the confirmation message
9. In the cafeteria form press the MY ORDERS button to view your order

Cleaning System

1. Press the CLEAN button
2. Choose cleaning areas, method and time
3. Press the START button
4. When the cleaning is finished, you will receive a message in the app regardless of what screen you are on

Map System

1. Press the MAP button
2. View your location, Bus location, Bus route and Bus stops
3. Click the road image to toggle between 3 different Driver's view angles.
4. Press any Landmark to see a description about it
5. Press Start Audio to hear the description, with options to Pause or Resume

Driving System (Driver Only)

1. Click the Menu Icon (Top Left)
2. Press the DRIVING MODE button
3. Assists in driving and notifies about speed, passengers exiting and doors status

Control Panel System(Passenger)

1. Click the Menu Icon (Top Left)
2. Press the CONTROL PANEL button
3. View A/C status and Temperature as well as Energy Information and Weather

Control Panel System(Employee)

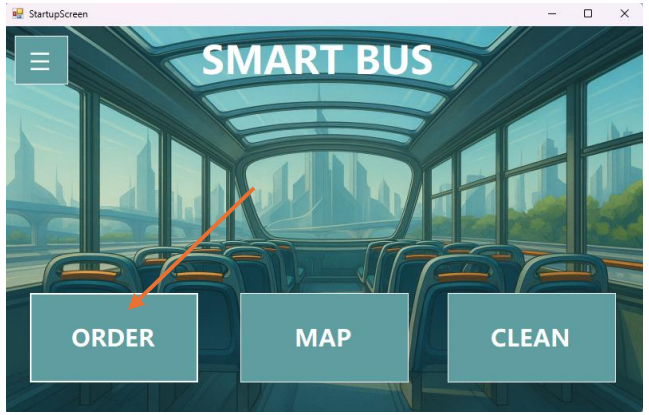
1. Click the Menu Icon (Top Left)
2. Press the CONTROL PANEL button
3. View weather and Energy Information
4. Change A/C temperature, A/C status and A/C mode
5. Change roof status manually

QUICK REFERENCE MANUAL

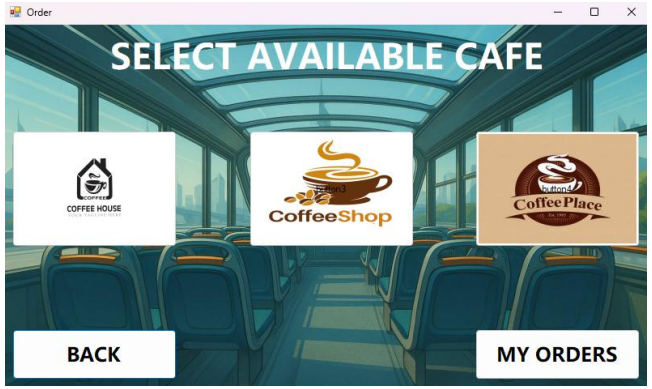
What is it About

- This manual presents the syntax of commands in a highly concise and summarized manner.
- It is designed for users who already know how to use the program in general terms.
- It serves as a fast lookup for users who may not remember the specific sequence or "syntax" of a particular command.
- Unlike the Quick Start notes, this section acts as a final "cheat sheet" for the most essential interactions across all systems.
- The goal is to provide maximum information with minimum reading for the Ordering, Cleaning, Map, Driving, and Control Panel systems.

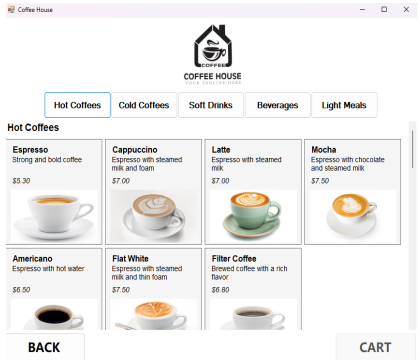
Ordering System



1. Press the ORDER button

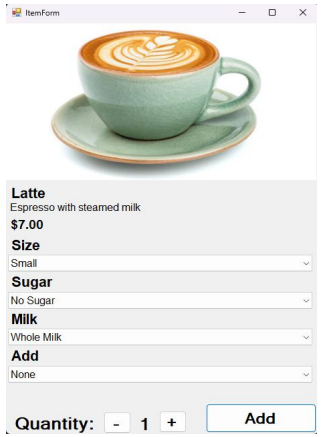


2. Choose your desired cafeteria



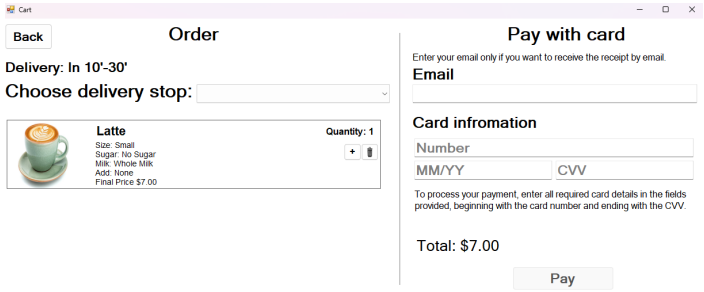
- Choose the items you want
- Press the category buttons to navigate through the categories quickly
- When finished press the CART button

3. Create order



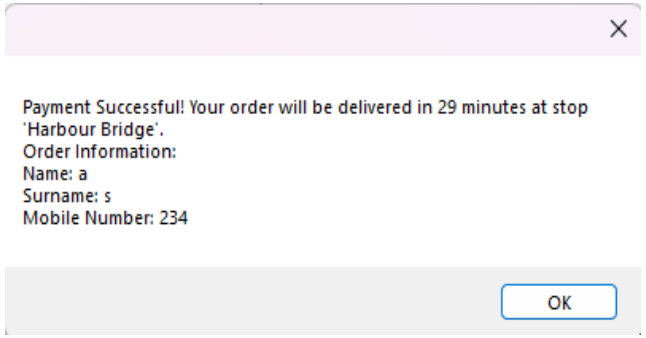
- When you click on an item, you can customize it like size, sugar etc.
- Adjust the quantity
- Press the ADD button

4. Customize your Item



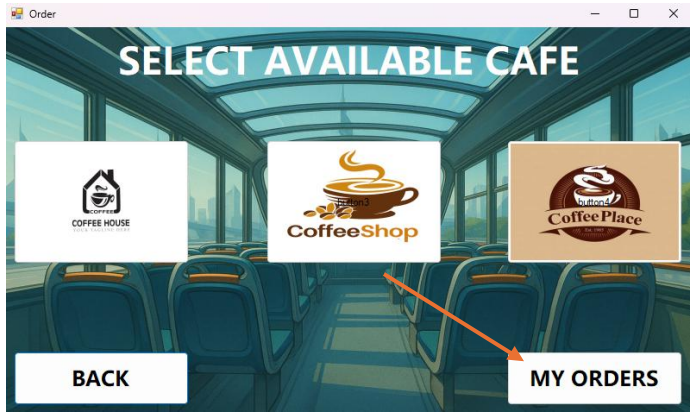
- View your items and choose delivery stop
- Fill your Card Information and email (optional) for receipt
- Press the PAY button

5. Payment



- You must see a confirmation message after paying

6. Confirmation

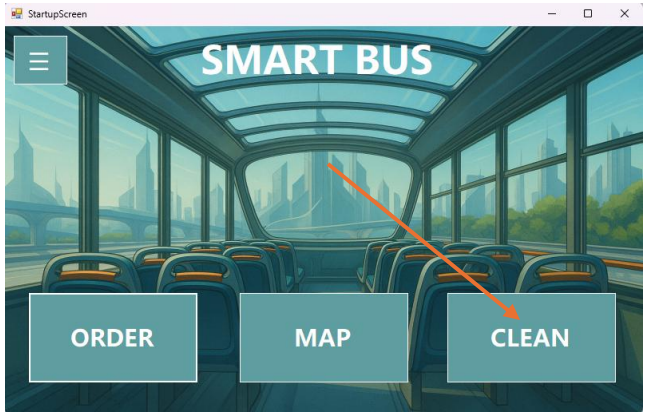


7. Press My orders

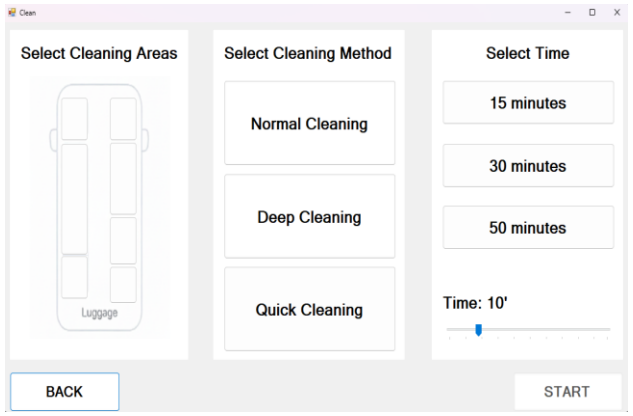
8. View your orders and details

9. Ordered Successfully

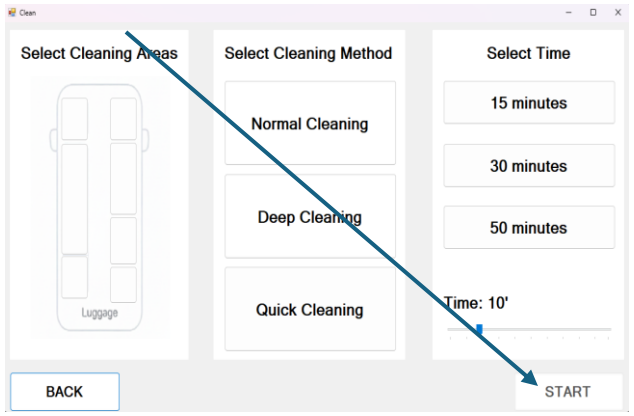
Cleaning System



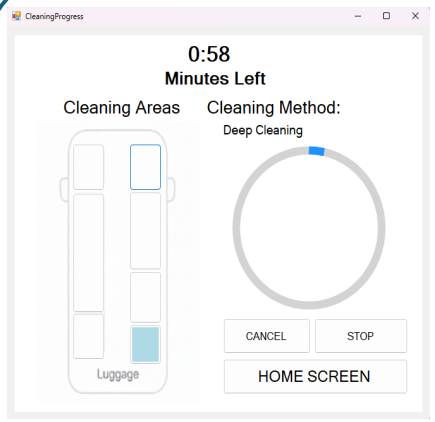
1. Press the CLEAN button



2. Choose Cleaning areas, method and Time

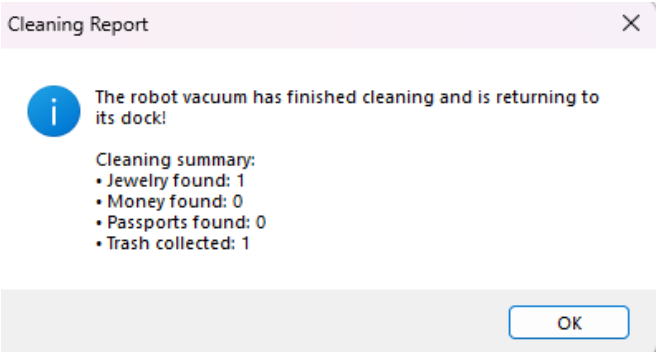


3. Press the START button



- You can view the progress of the cleaning
- Press CANCEL, STOP to stop or cancel the cleaning
- Press HOME SCREEN button to see other app features

4. Cleaning Progress

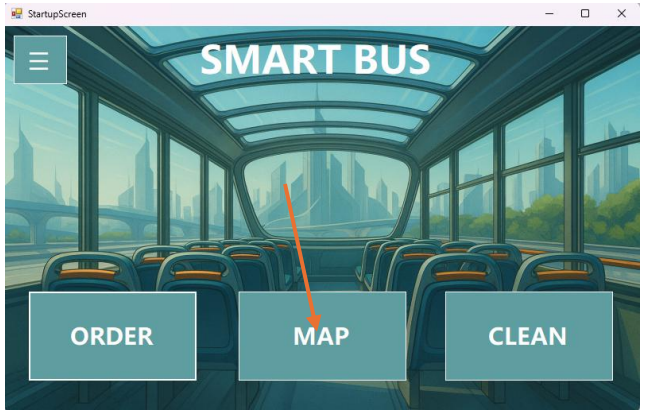


- Cleaning report after cleaning is completed

5. View Cleaning Report

6. Cleaned Successfully

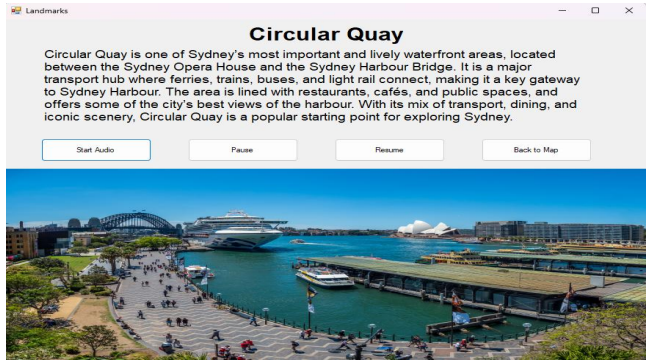
Map System



1. Press the MAP button

- View your location, bus location and bust stops
- Choose a landmark or driver's view

2. Map Navigation



- Text information or Audio
- Pause and Resume buttons to not miss anything

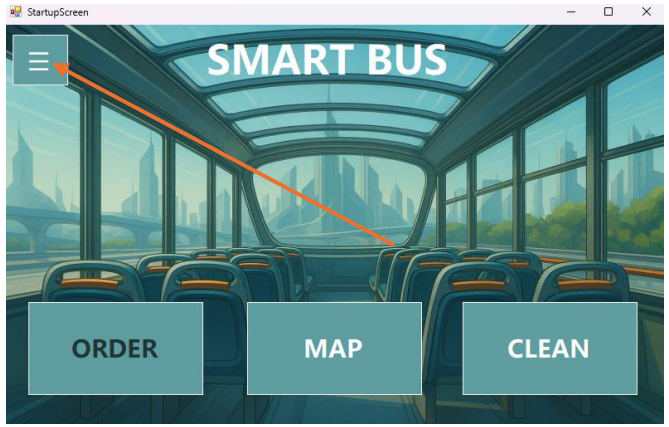
3. Landmark Information

- Swipe to switch between 3 different camera angles of the driver

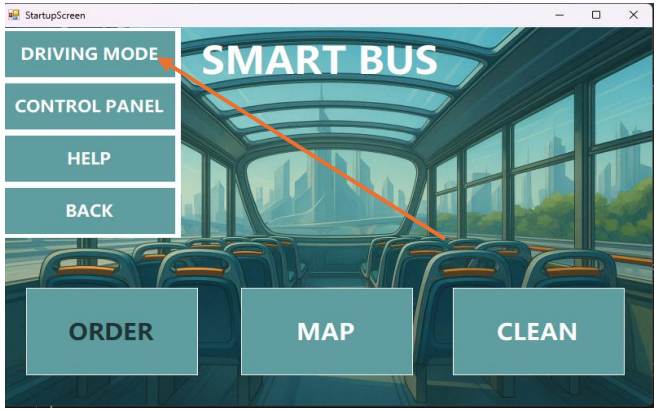
4. Driver's View

5. Map Opened Successfully

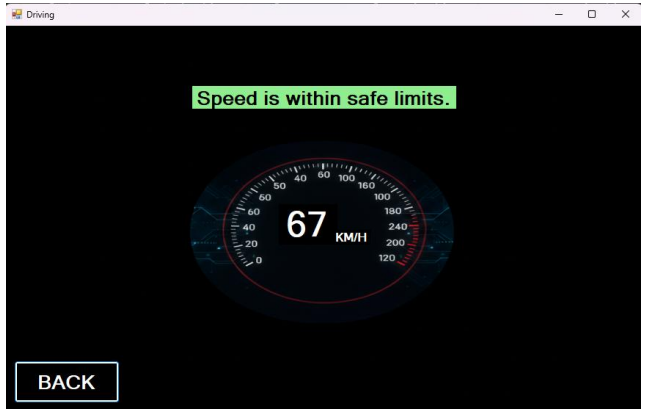
Driving System (Driver Only)



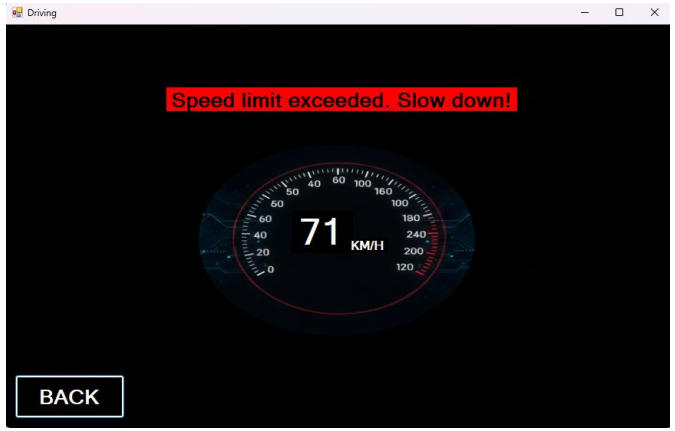
1. Press the Menu Icon button (Top Left)



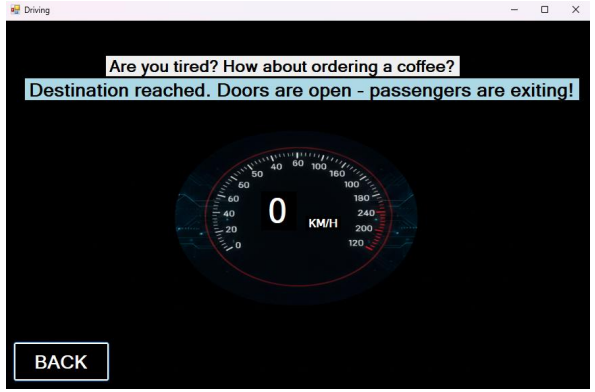
2. Press the DRIVING MODE button



3. Assist in driving with messages for speed



4. Speed limit exceeded, message and sound

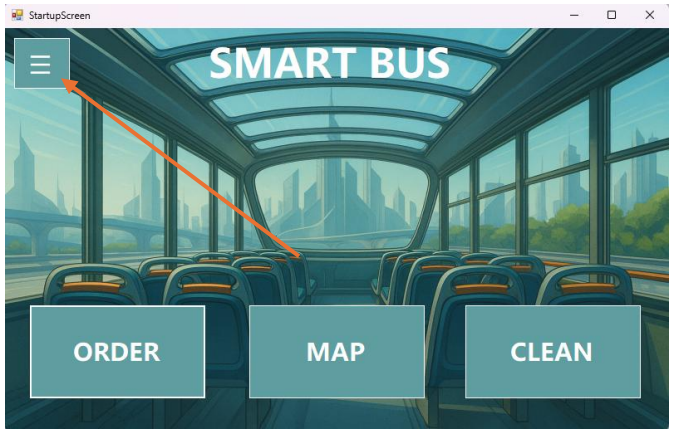


- Asking for coffee and updating on door status

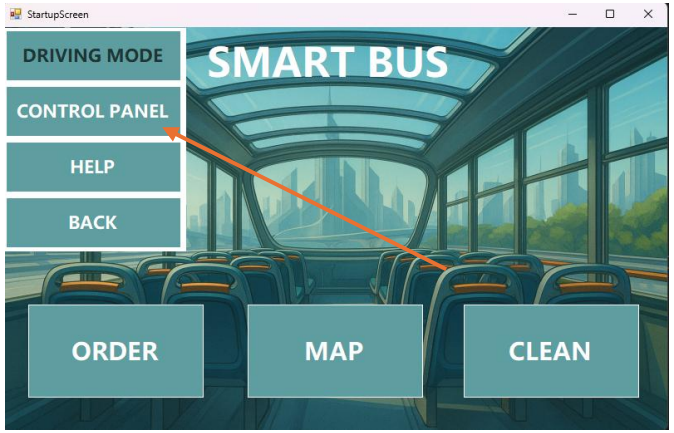
5. At bus stop

6. Driving Mode Opened Successfully

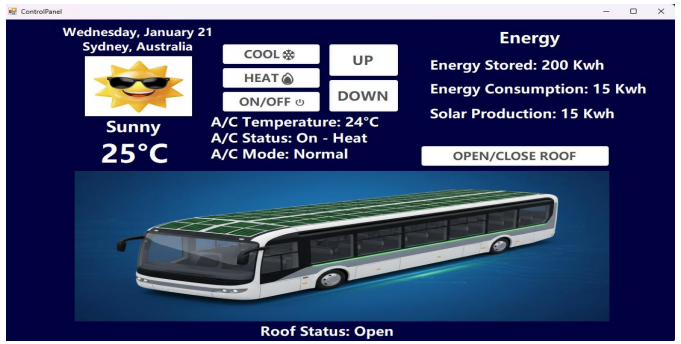
Control Panel System (Passenger)



1. Press the Menu Icon button (Top Left)



2. Press the CONTROL PANEL button

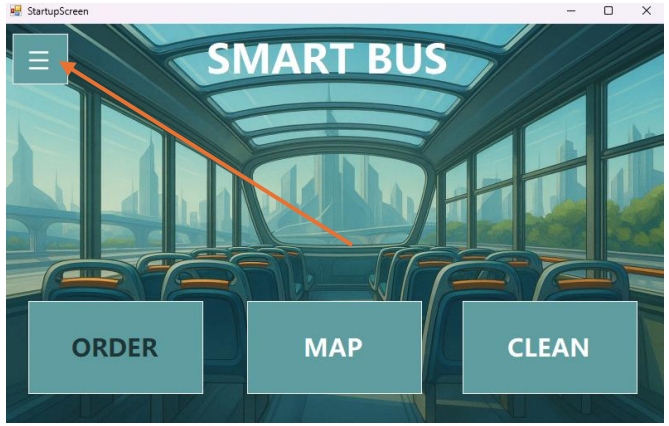


- Information about A/C, Weather, roof status and Energy

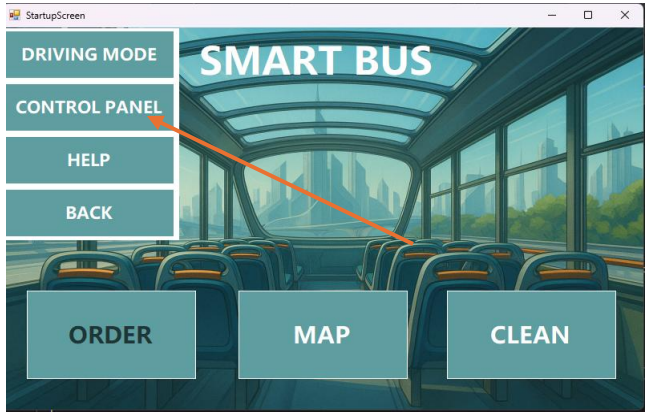
3. Control Panel Navigation

4. Control panel (Passenger) Opened Successfully

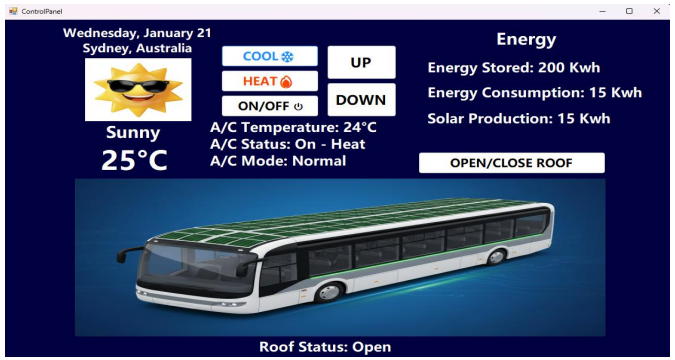
Control Panel System (Employee)



1. Press the Menu Icon button (Top Left)

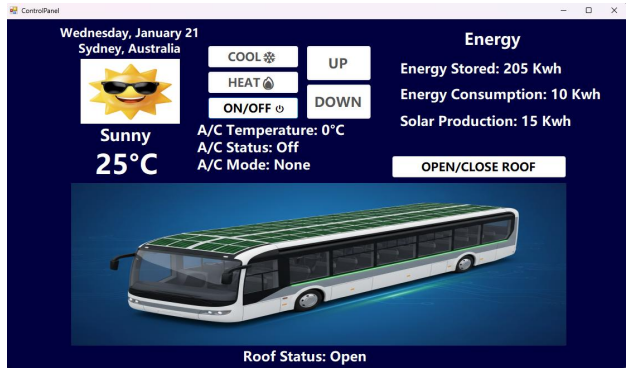


2. Press the CONTROL PANEL button



- Information about A/C, Weather, roof status and Energy

3. Control Panel Navigation



- Change A/C mode, status and Temperature
- Open/Close roof manually

4. Employee Operations in Control Panel

5. Control panel (Employee) Opened Successfully

DETAILED SYSTEM INSTRUCTIONS

What is it About

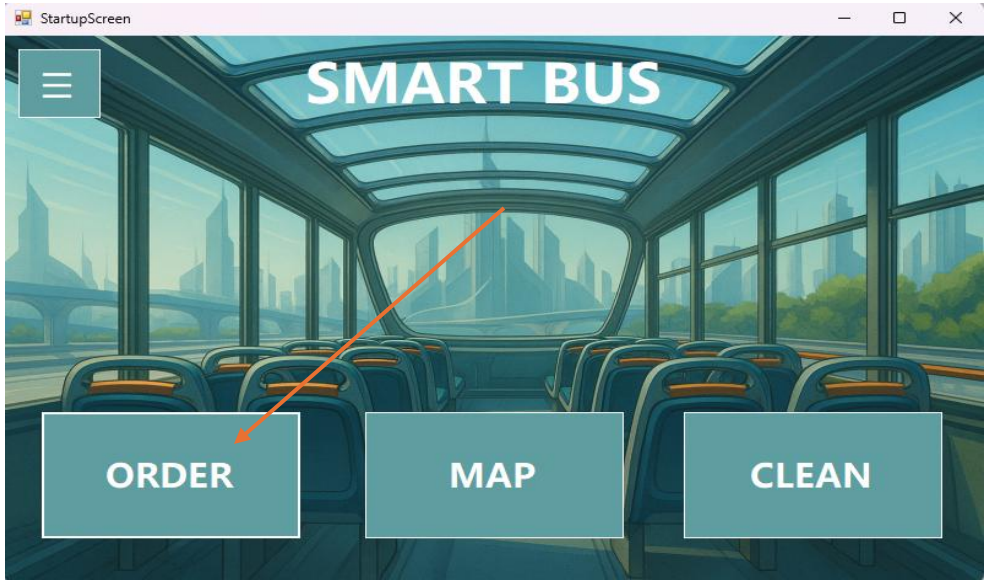
- This section provides a comprehensive, step-by-step guide for every system within the Smart Bus application.
- It covers both Passenger functionalities, such as the Ordering, Cleaning and Map systems, and Employee/Driver operations, including Driving and the Control Panel.
- These instructions are designed to lead the user through the exact visual flow of the app, from the initial Menu selection to the final Success confirmation.
- Each task is broken down into numbered actions to ensure that complex features are simple and easy to execute.

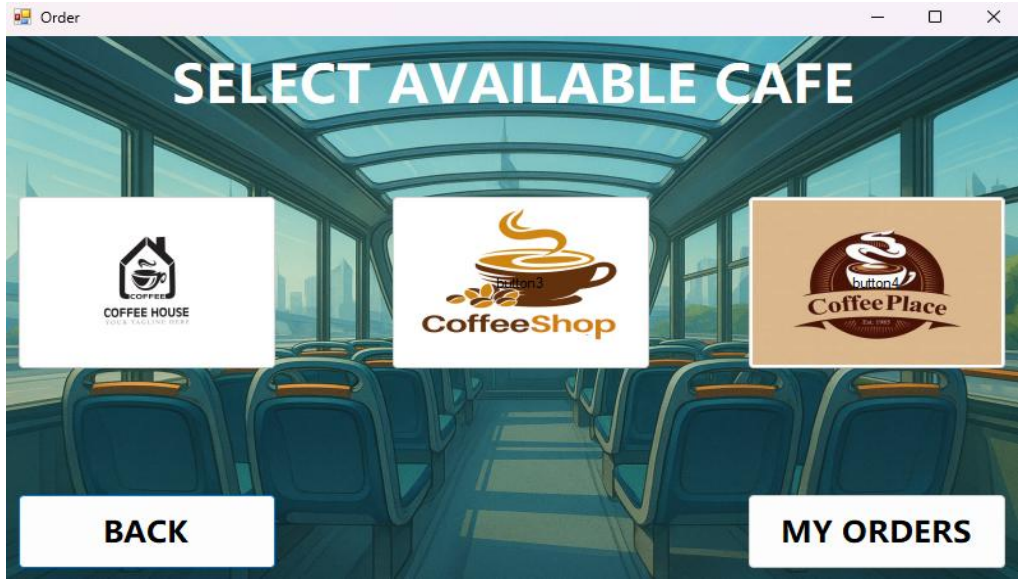
ORDERING SYSTEM

- This system allows passengers to connect with partner cafeterias to order coffee, drinks, and light meals. Order delivery takes place at the specific bus stop chosen by the user. Payment is accepted exclusively via card.

Choose Cafeteria

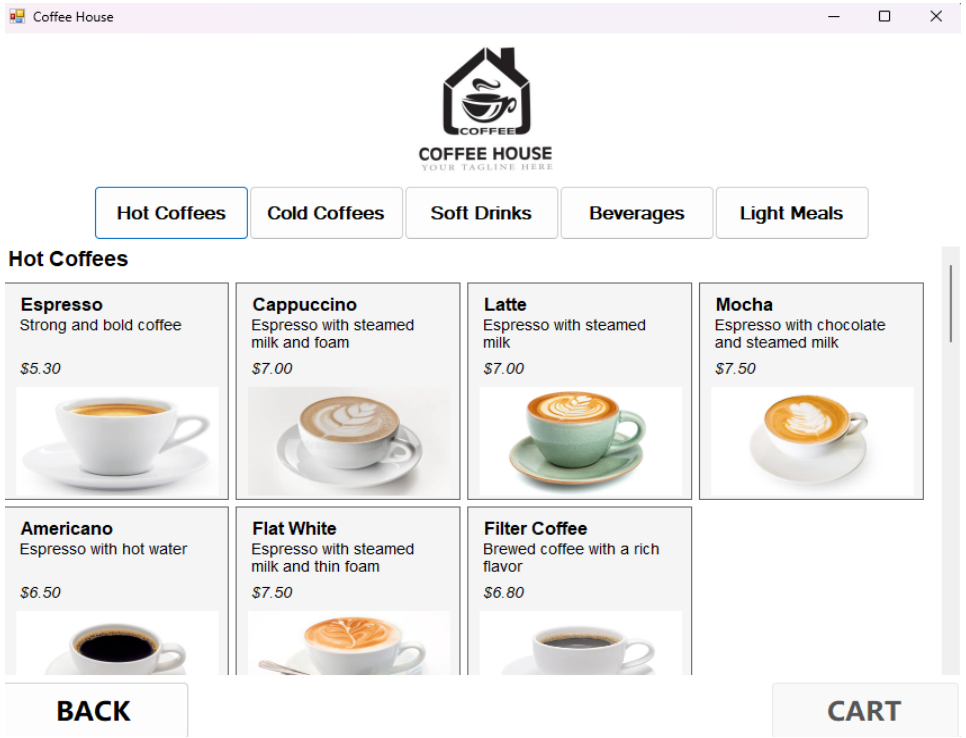
1. From the main Screen, press the large ORDER button to enter the catering interface.
2. You will be presented with a list of available local vendors, such as Coffee House, CoffeeShop, and Coffee Place.
3. Simply tap the logo of your desired cafeteria to view their specific menu.

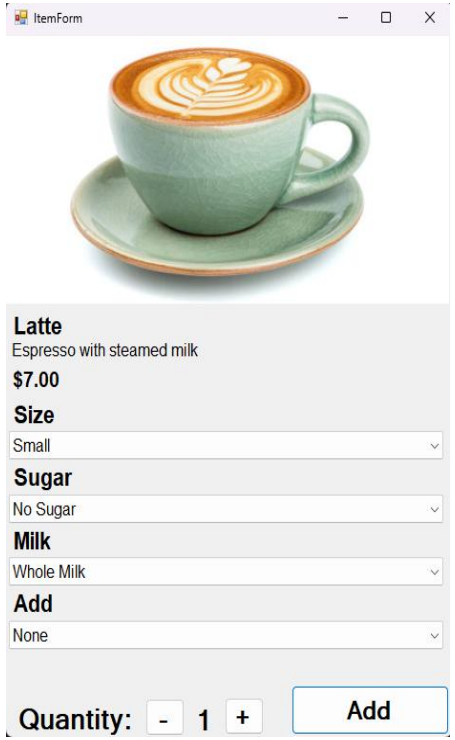




Add Items

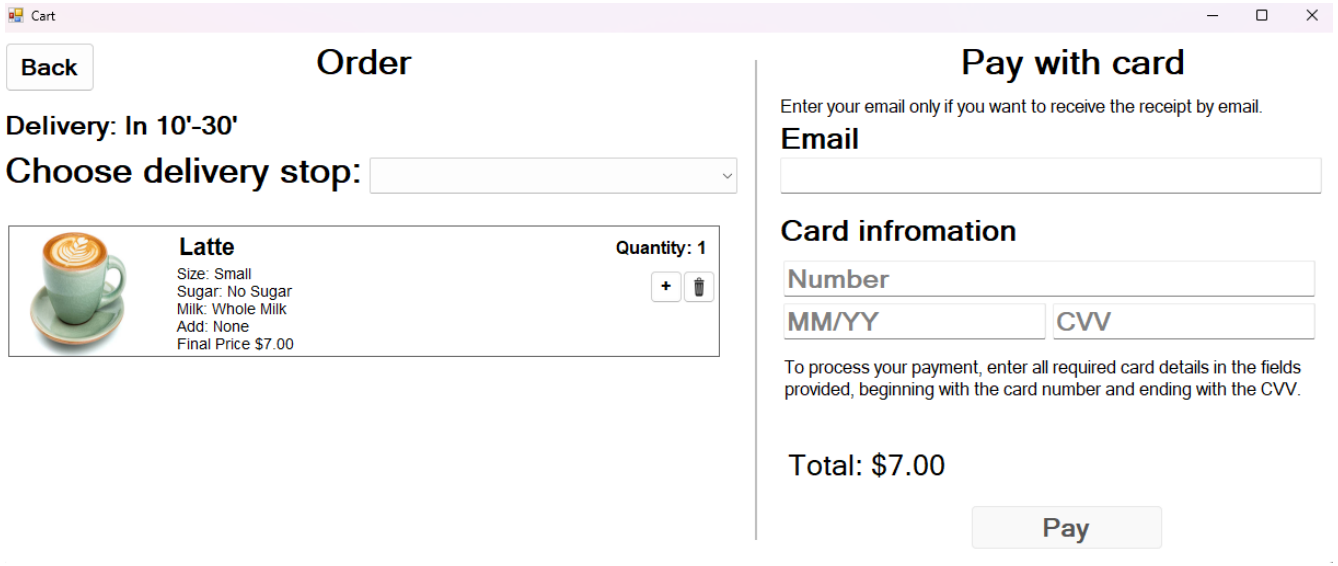
1. Browse the categories at the top of the menu by clicking or just scrolling, such as Hot Coffees, Soft Drinks, Light Meals, etc. to find your item.
2. Click on a specific product, like a Latte or Mocha, to open the ItemForm customization window.
3. Use the dropdown menus to select your preferred Size, Sugar, Milk, etc. options.
4. Adjust the Quantity using the + and - buttons and then press Add to place the item into your virtual cart.

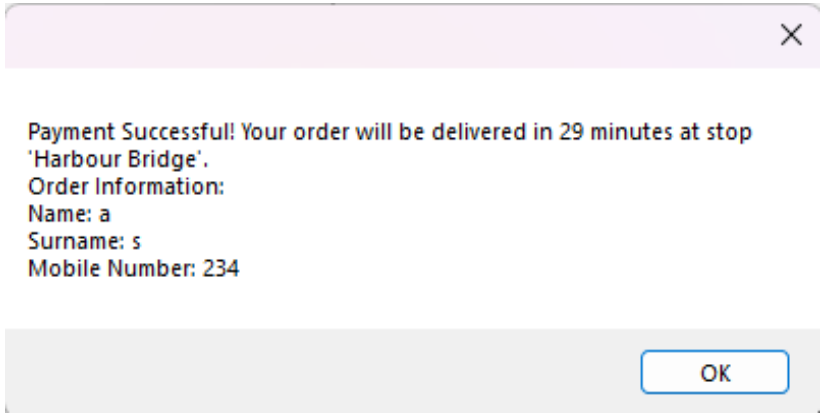




Cart & Payment

1. Once your selection is complete, press the CART button to proceed to the checkout screen.
2. Use the Choose delivery stop dropdown to select the bus stop where you want your order delivered.
3. In the Pay with card section, enter your Email (optional for a digital receipt) and your Card information, including card number, expiry date, and CVV.
4. Press the Pay button to finalize the transaction.
5. Receive a confirmation message right after paying successfully with your delivery time, bus stop and order information.





View Orders

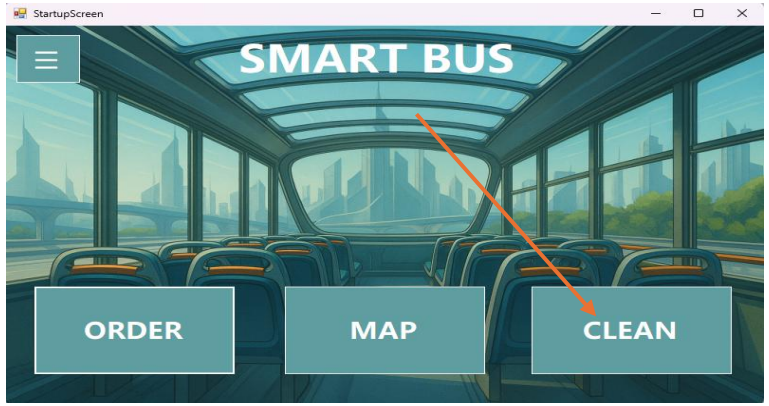
1. To check your current orders, return to the cafeteria selection screen and press the MY ORDERS button.
2. The MyOrder screen allows you to review the full details (except your personal information for safety reasons) of your purchase, including the items ordered, the total price, the expected delivery window and the bus stop you have chosen.

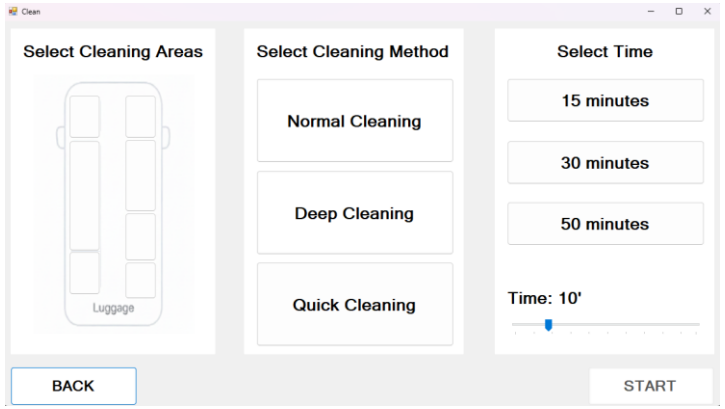
CLEANING SYSTEM

- This system provides full control over the bus's autonomous robot vacuum. Users can select specific cleaning areas, choose the cleaning method, and set the duration of the session. Additionally, the robot uses its integrated camera to identify trash or lost items, notifying the company whenever a valuable object is found.

Choose Areas, Method & Time

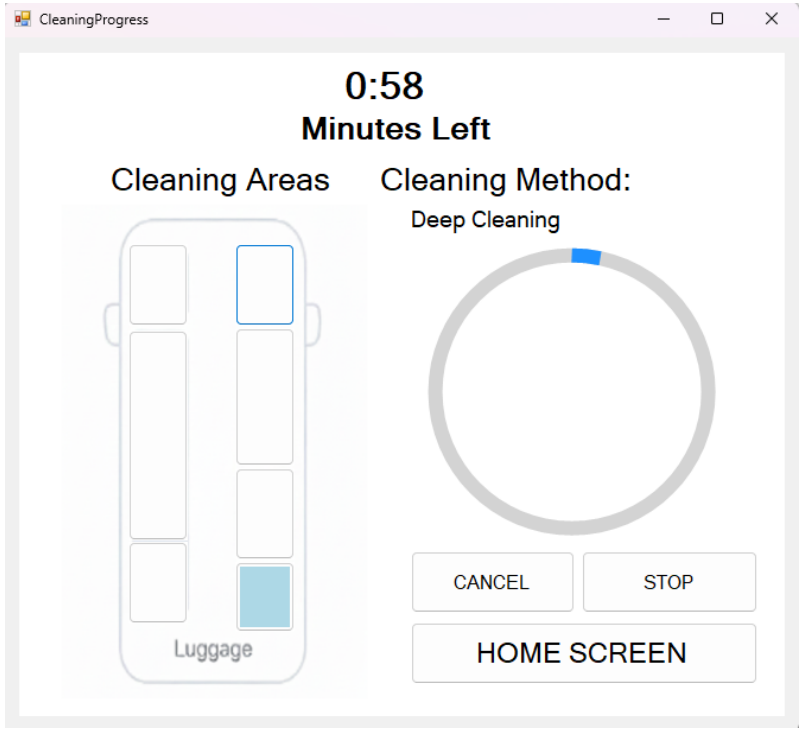
1. To begin, press the CLEAN button on the main Startup Screen.
2. Once the cleaning interface opens, you can select the areas you want the robot to focus on by tapping the corresponding sections on the bus floor plan.
3. Next, choose your preferred cleaning method: Normal, Deep, or Quick Cleaning.
4. Finally, set the duration for the session by selecting one of the preset time buttons (15, 30, or 50 minutes) or by using the manual slider.

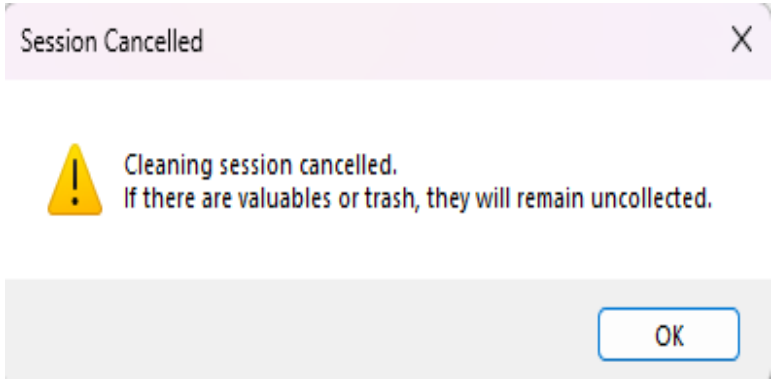




Start Cleaning & Watch Progress

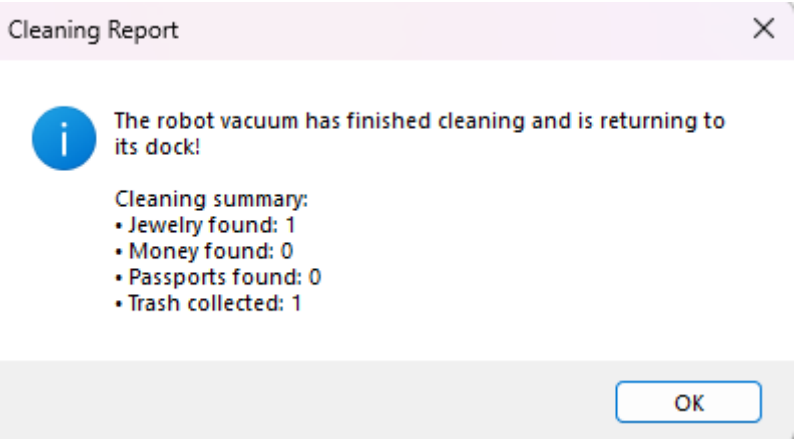
1. After configuring your settings, press the START button.
2. This will take you to the progress screen where you can see a live countdown of the minutes remaining and a visual map of the robot's cleaning areas that you chose.
3. While the robot is active, you can press HOME SCREEN to use other features of the app, the robot will continue its work in the background.
4. If you need to end the session early, you can use the CANCEL or STOP buttons, though a warning will appear noting that trash or valuables will remain uncollected.





Cleaning Report

1. When the robot vacuum finishes its task, it automatically returns to its charging dock.
2. A Cleaning Report will then appear on the screen to provide a final summary of the session.
3. This report includes a list of any lost items found, such as jewelry, money, or passports, as well as a count of the trash items collected.
4. Once you review the report just press OK.

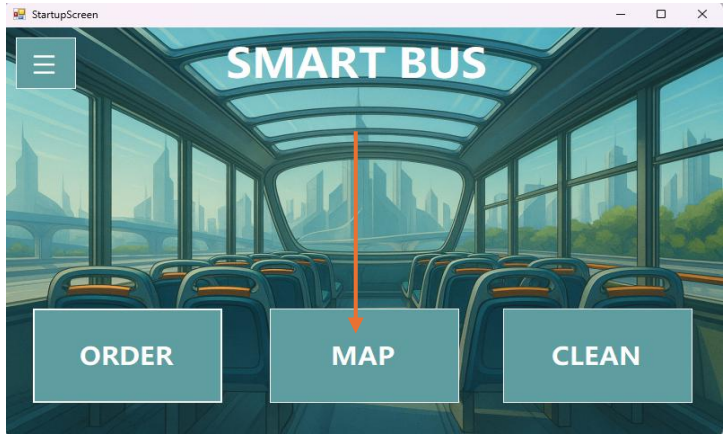


MAP SYSTEM

- The Map System provides passengers with a real-time, interactive navigation experience. It allows users to track their personal location and the bus's live position along a designated route. Passengers can identify upcoming bus stops and access multimedia information about significant landmarks throughout the city. Additionally, the system features a unique driver-view perspective, offering three different camera angles for an immersive journey.

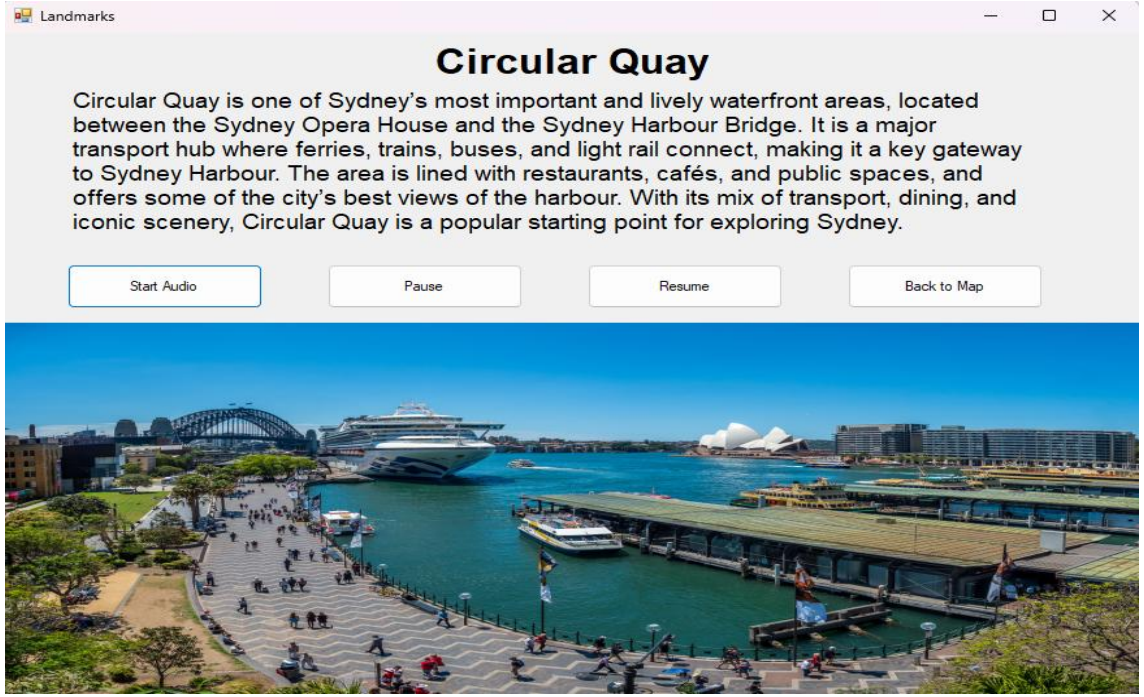
Bus Route, Bus Stops & Locations

1. To enter the navigation interface, select the MAP button from the Main Screen.
2. The map displays the full bus route and highlights your current location alongside the live location of the bus.
3. Red markers on the map indicate scheduled bus stops, and landmarks like the Sydney Opera House are marked with specific icons for easy identification.



View Landmark Information

1. Users can explore city landmarks by clicking directly on a map icon.
2. Choosing a landmark opens a detailed information page containing a historical description and a high-resolution image.
3. You can also listen to this information by pressing Start Audio.
4. For full control, the system includes Pause and Resume buttons so you can manage the audio guide at your own pace.
5. Whenever you want just press the Back to map button to go back.



Driver’s View

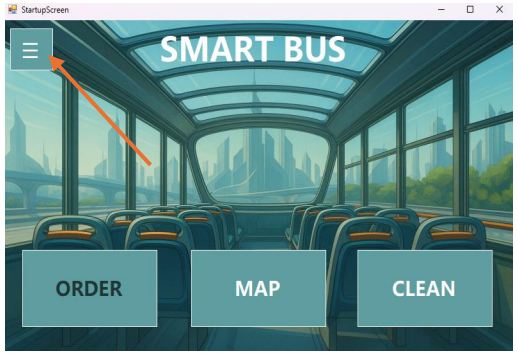
1. To see the road from the driver's perspective, click the driver's view preview image on the main map.
2. This feed provides a live look at the path ahead.
3. You can swipe left or right across the view to switch between three different camera angles, allowing you to see exactly what the driver or the autonomous system sees during transit.

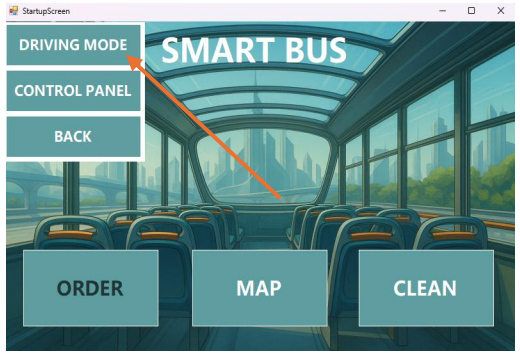
DRIVING SYSTEM (DRIVER ONLY)

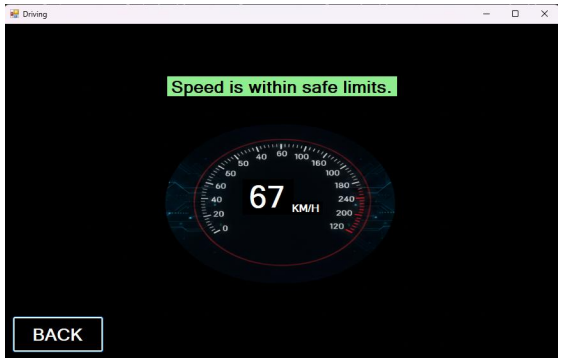
- The Driving System is a specialized interface designed for the bus driver to ensure a safe and efficient journey. This real-time monitoring tool provides critical safety alerts regarding speed limits and gives status updates on passenger activity at every bus stop. The system also includes assistive features that alert the driver in the event of lane-drifting or signs of fatigue.

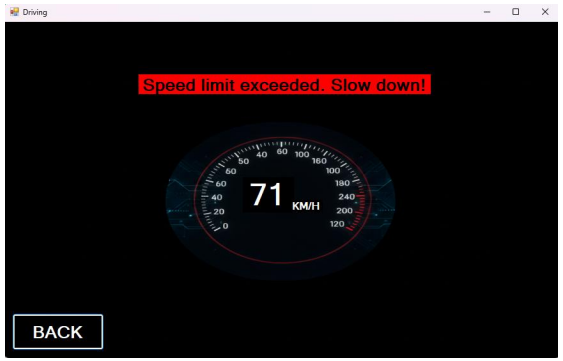
Assist in Driving and Speed Limit Alerts

1. To access the driving system, the driver must first press the Menu Icon button in the top-left corner of the screen and select DRIVING MODE.
2. Once the system is active, it continuously monitors the bus's speed.
3. When traveling at a safe pace , the interface displays a green message confirming that the speed is within safe limits.
4. If the bus exceeds the speed limit the system triggers a prominent red warning message and an alert sound to prompt the driver to slow down immediately.



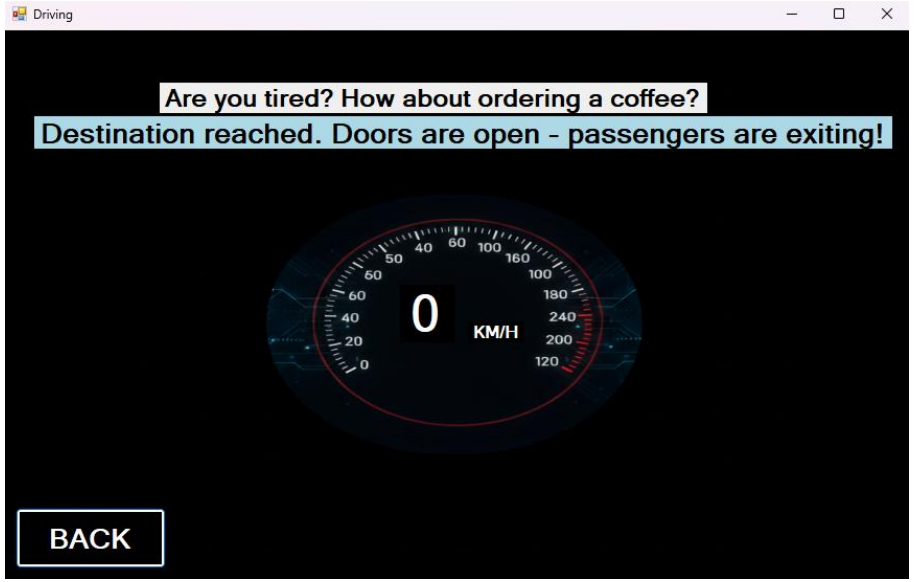






Bus Stop Messages

1. The system provides automated updates based on the bus's location and mechanical status at every scheduled stop.
2. Upon reaching a destination, the driver is notified with a live status message, such as "Destination reached. Doors are open - passengers are exiting!" to ensure they remain aware of passenger movement while the bus is stationary.
3. Additionally, the system prioritizes driver wellness by monitoring for signs of exhaustion, if the driver appears tired, the system will proactively suggest taking a break by asking, "Are you tired? How about ordering a coffee?".

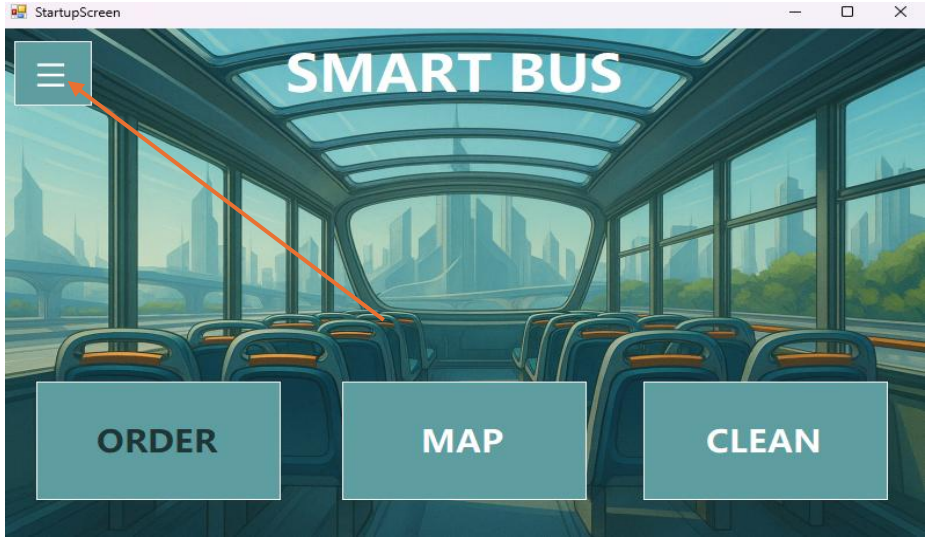


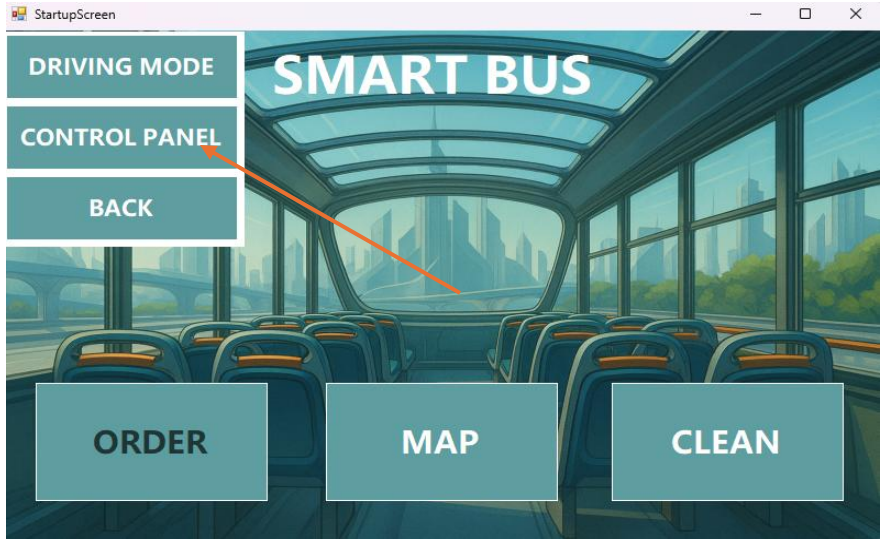
CONTROL PANEL SYSTEM

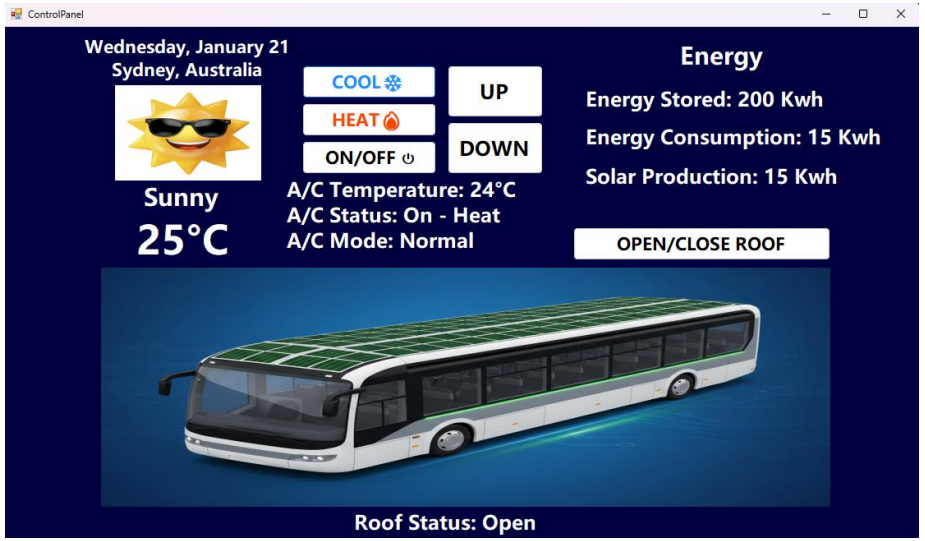
- The Control Panel System is a dual-purpose interface designed for both passengers and employees to monitor and manage the bus's internal environment. While passengers can view real-time data regarding weather, energy usage, and current climate settings, employees are granted administrative access to actively adjust the A/C modes and manually control the bus's panoramic roof. The system can also automatically close the roof if the weather is bad and can suggest energy-saving tips to employees.

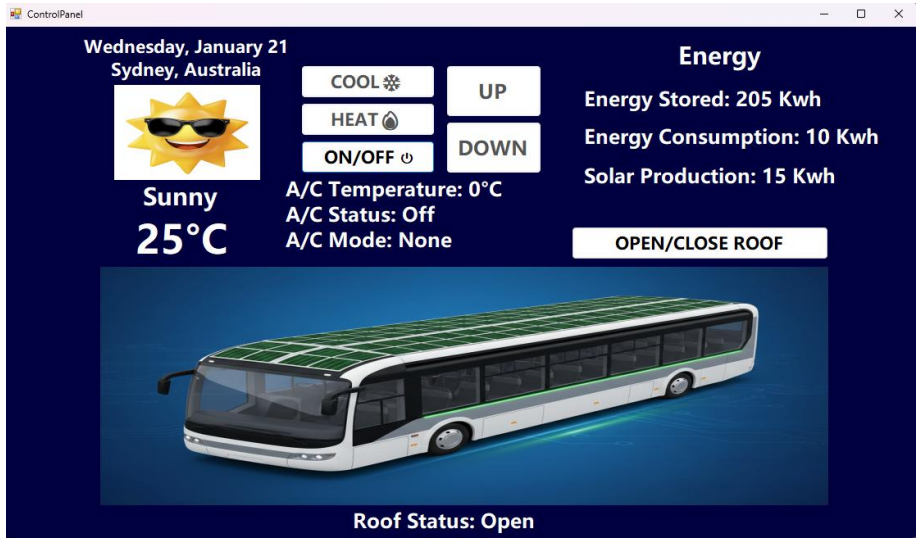
Choose A/C Mode, Temperature & Open/Close Roof (Employees Only)

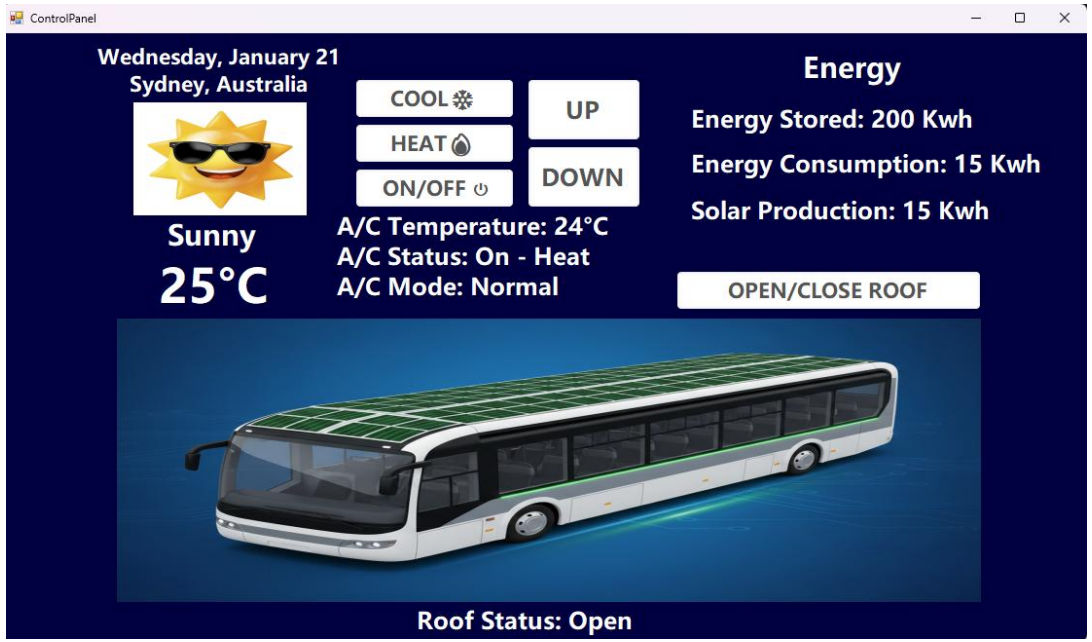
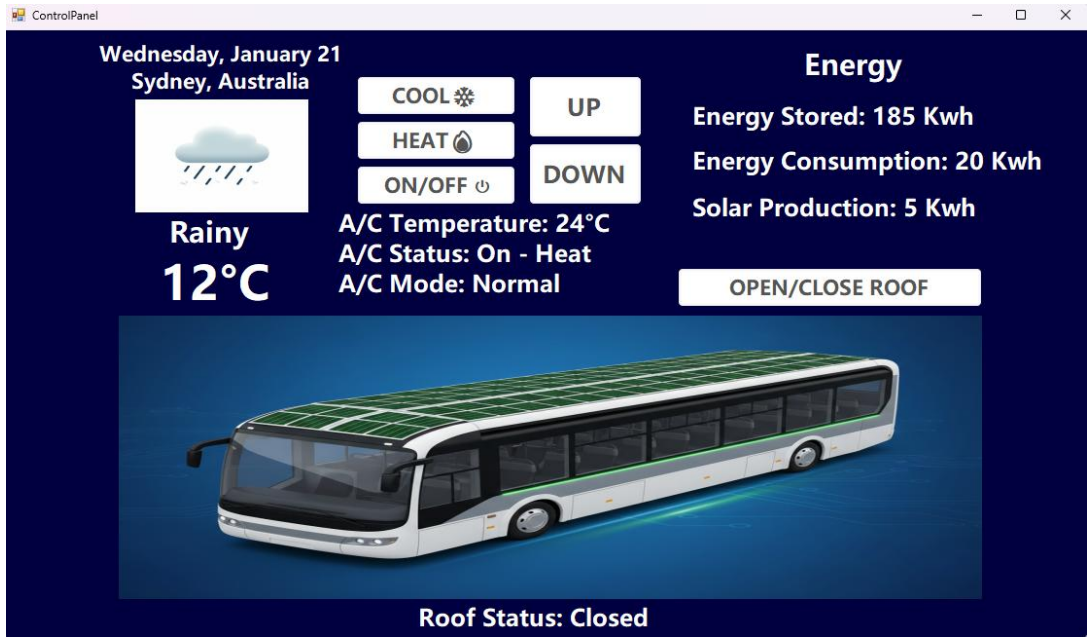
1. To manage the bus environment, an employee must press the Menu Icon button and select CONTROL PANEL.
2. In this active mode, the employee can toggle the A/C system ON or OFF, switch between COOL and HEAT modes, and use the UP and DOWN buttons to set the specific bus temperature.
3. Additionally, the employee has the exclusive ability to press the OPEN/CLOSE ROOF button, which manually operates the bus's roof based on passenger needs or weather conditions.
4. The employee can also like the passenger view the energy information like energy stored, consumption and solar production.


StartupScreen
SMART BUS
ORDER
MAP
CLEAN


StartupScreen
DRIVING MODE
CONTROL PANEL
BACK
SMART BUS
ORDER
MAP
CLEAN


ControlPanel
Wednesday, January 21
Sydney, Australia
Sunny
25°C
COOL
HEAT
ON/OFF
UP
DOWN
A/C Temperature: 24°C
A/C Status: On - Heat
A/C Mode: Normal
Energy
Energy Stored: 200 Kwh
Energy Consumption: 15 Kwh
Solar Production: 15 Kwh
OPEN/CLOSE ROOF
Roof Status: Open


ControlPanel
Wednesday, January 21
Sydney, Australia
Sunny
25°C
COOL
HEAT
ON/OFF
UP
DOWN
A/C Temperature: 0°C
A/C Status: Off
A/C Mode: None
Energy
Energy Stored: 205 Kwh
Energy Consumption: 10 Kwh
Solar Production: 15 Kwh
OPEN/CLOSE ROOF
Roof Status: Open

View Weather, Energy & A/C Mode (Passenger)

1. The Control Panel provides a live Energy summary, showing the amount of Energy Stored, current Energy Consumption, and Solar Production generated by the bus's roof panels.
2. It also displays the current date, time, and local weather—such as whether it is Sunny or Rainy—along with the current A/C Status and Roof Status.